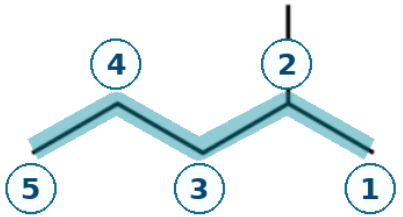
CARD A01 — STRUCTURE → NAME

FOUNDATION • Molecular formula: C6H14 • Structure from instructor PDF

01 / NUMBERED PARENT



02 / NAMING TRAIL

PARENT 5-carbon chain → pentane

NUMBERING From left: methyl C4; from right: C2. Choose C2.

SUBSTITUENTS One methyl at C2.

FOCUS Find longest chain; number from end nearest branch.

03 / WORKED IUPAC NAME (VERIFY WITH YOUR OWN TRAIL)

2-methylpentane

04 / THE ALKANE CHECK

EXPECTED IUPAC NAME: 2-methylpentane

The Alkane result: NOT YET CHECKED.

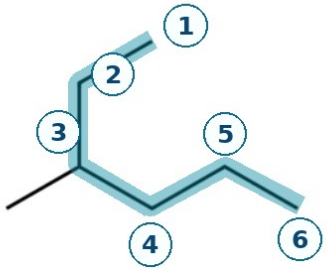
Insert genuine cropped tool screenshot after rebuilding the structure.

✓ parent checked ✓ locants checked ✓ formula counted ✓ name punctuated ☐ website check pending

CARD A02 — STRUCTURE → NAME

FOUNDATION • Molecular formula: C7H16 • Structure from instructor PDF

01 / NUMBERED PARENT



02 / NAMING TRAIL

PARENT 6-carbon bent chain → hexane

NUMBERING From upper end: C3; from right end: C4. Choose C3.

SUBSTITUENTS One methyl at C3.

FOCUS Longest chain bends; do not assume the horizontal line is longest.

03 / WORKED IUPAC NAME (VERIFY WITH YOUR OWN TRAIL)

3-methylhexane

04 / THE ALKANE CHECK

EXPECTED IUPAC NAME: 3-methylhexane

The Alkane result: NOT YET CHECKED.

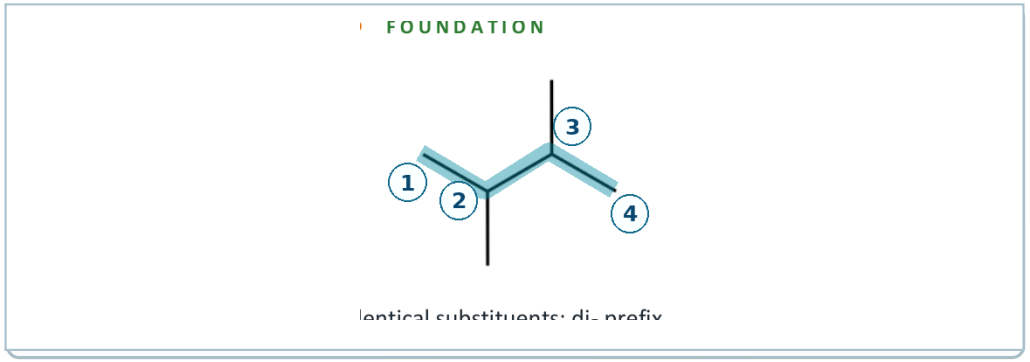
Insert genuine cropped tool screenshot after rebuilding the structure.

✓ parent checked ✓ locants checked ✓ formula counted ✓ name punctuated ☐ website check pending

CARD A03 — STRUCTURE → NAME

FOUNDATION • Molecular formula: C6H14 • Structure from instructor PDF

01 / NUMBERED PARENT



02 / NAMING TRAIL

PARENT 4-carbon chain → butane

NUMBERING Both directions give 2,3 (a symmetry tie).

SUBSTITUENTS Methyl at C2 and C3; use the di- prefix.

FOCUS Use di- and include both locants.

03 / WORKED IUPAC NAME (VERIFY WITH YOUR OWN TRAIL)

2,3-dimethylbutane

04 / THE ALKANE CHECK

EXPECTED IUPAC NAME: 2,3-dimethylbutane

The Alkane result: NOT YET CHECKED.

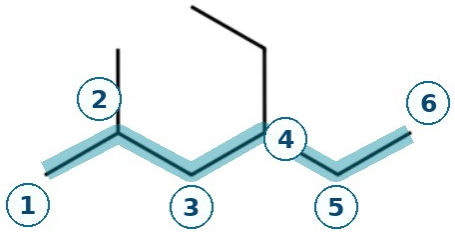
Insert genuine cropped tool screenshot after rebuilding the structure.

✓ parent checked ✓ locants checked ✓ formula counted ✓ name punctuated ☐ website check pending

CARD A04 — STRUCTURE → NAME

DEVELOPING • Molecular formula: C9H20 • Structure from instructor PDF

01 / NUMBERED PARENT



02 / NAMING TRAIL

PARENT 6-carbon chain → hexane

NUMBERING Left to right: 2,4; right to left: 3,5. Choose 2,4.

SUBSTITUENTS Ethyl at C4; methyl at C2 (alphabetical order).

FOCUS Alphabetical citation order does not follow locant order.

03 / WORKED IUPAC NAME (VERIFY WITH YOUR OWN TRAIL)

4-ethyl-2-methylhexane

04 / THE ALKANE CHECK

EXPECTED IUPAC NAME: 4-ethyl-2-methylhexane

The Alkane result: NOT YET CHECKED.

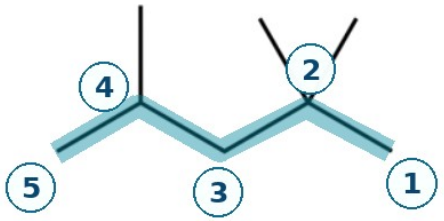
Insert genuine cropped tool screenshot after rebuilding the structure.

✓ parent checked ✓ locants checked ✓ formula counted ✓ name punctuated ☐ website check pending

CARD A05 — STRUCTURE → NAME

DEVELOPING • Molecular formula: C8H18 • Structure from instructor PDF

01 / NUMBERED PARENT



02 / NAMING TRAIL

PARENT 5-carbon chain → pentane

NUMBERING Left to right: 2,4,4; right to left: 2,2,4.

SUBSTITUENTS Two methyls at C2; one methyl at C4.

FOCUS Compare numbering lists at first point of difference.

03 / WORKED IUPAC NAME (VERIFY WITH YOUR OWN TRAIL)

2,2,4-trimethylpentane

04 / THE ALKANE CHECK

EXPECTED IUPAC NAME: 2,2,4-trimethylpentane

The Alkane result: NOT YET CHECKED.

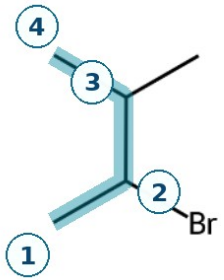
Insert genuine cropped tool screenshot after rebuilding the structure.

✓ parent checked ✓ locants checked ✓ formula counted ✓ name punctuated ☐ website check pending

CARD A06 — STRUCTURE → NAME

DEVELOPING • Molecular formula: C5H11Br • Structure from instructor PDF

01 / NUMBERED PARENT



02 / NAMING TRAIL

PARENT	4-carbon chain → butane
NUMBERING	Both ends give 2,3; alphabetical tie: bromo gets C2.
SUBSTITUENTS	Bromo at C2; methyl at C3.

FOCUS Treat the halogen as a prefix; resolve ties alphabetically.

03 / WORKED IUPAC NAME (VERIFY WITH YOUR OWN TRAIL)

2-bromo-3-methylbutane

04 / THE ALKANE CHECK

EXPECTED IUPAC NAME: 2-bromo-3-methylbutane

The Alkane result: NOT YET CHECKED.

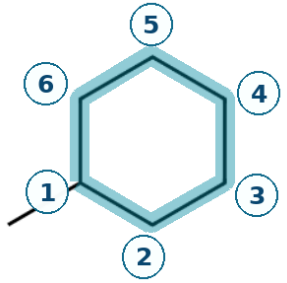
Insert genuine cropped tool screenshot after rebuilding the structure.

✓ parent checked ✓ locants checked ✓ formula counted ✓ name punctuated ☐ website check pending

CARD A07 — STRUCTURE → NAME

DEVELOPING • Molecular formula: C7H14 • Structure from instructor PDF

01 / NUMBERED PARENT



02 / NAMING TRAIL

PARENT 6-carbon ring → cyclohexane

NUMBERING One ring substituent: a locant is unnecessary.

SUBSTITUENTS One methyl on the cyclohexane ring.

FOCUS Use the ring as parent; a lone substituent needs no locant.

03 / WORKED IUPAC NAME (VERIFY WITH YOUR OWN TRAIL)

methylcyclohexane

04 / THE ALKANE CHECK

EXPECTED IUPAC NAME: methylcyclohexane

The Alkane result: NOT YET CHECKED.

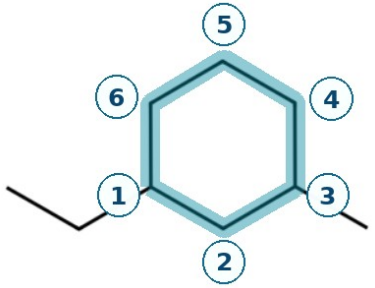
Insert genuine cropped tool screenshot after rebuilding the structure.

✓ parent checked ✓ locants checked ✓ formula counted ✓ name punctuated ☐ website check pending

CARD A08 — STRUCTURE → NAME

CHALLENGE • Molecular formula: C9H18 • Structure from instructor PDF

01 / NUMBERED PARENT



02 / NAMING TRAIL

PARENT	6-carbon ring → cyclohexane
NUMBERING	Ethyl as C1: nearest methyl at C3 (not C5).
SUBSTITUENTS	Ethyl at C1; methyl at C3; ethyl alphabetically first.

FOCUS Number in both directions, use alphabetical tie-break.

03 / WORKED IUPAC NAME (VERIFY WITH YOUR OWN TRAIL)

1-ethyl-3-methylcyclohexane

04 / THE ALKANE CHECK

EXPECTED IUPAC NAME: 1-ethyl-3-methylcyclohexane

The Alkane result: NOT YET CHECKED.

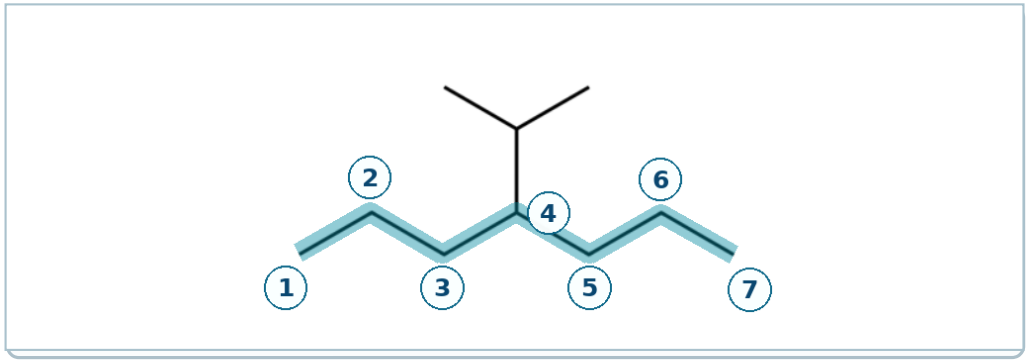
Insert genuine cropped tool screenshot after rebuilding the structure.

✓ parent checked ✓ locants checked ✓ formula counted ✓ name punctuated ☐ website check pending

CARD A09 — STRUCTURE → NAME

CHALLENGE • Molecular formula: C10H22 • Structure from instructor PDF

01 / NUMBERED PARENT



02 / NAMING TRAIL

PARENT 7-carbon horizontal chain → heptane

NUMBERING Either direction locates the substituent at C4.

SUBSTITUENTS Branched (1-methylethyl) at C4; also isopropyl.

FOCUS Trace longest chain through every branch point.

03 / WORKED IUPAC NAME (VERIFY WITH YOUR OWN TRAIL)

4-(1-methylethyl)heptane

04 / THE ALKANE CHECK

EXPECTED IUPAC NAME: 4-(1-methylethyl)heptane

The Alkane result: NOT YET CHECKED.

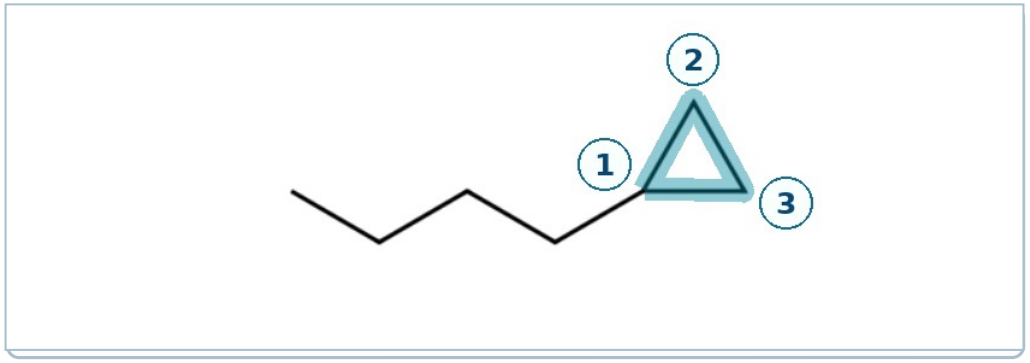
Insert genuine cropped tool screenshot after rebuilding the structure.

✓ parent checked ✓ locants checked ✓ formula counted ✓ name punctuated ☐ website check pending

CARD A10 — STRUCTURE → NAME

CHALLENGE • Molecular formula: C7H14 • Structure from instructor PDF

01 / NUMBERED PARENT



02 / NAMING TRAIL

PARENT 3-carbon ring → cyclopropane

NUMBERING Ring is the parent here, even beside a 4-C chain.

SUBSTITUENTS One straight-chain butyl group; ring locant omitted.

FOCUS For this activity ring takes precedence as parent.

03 / WORKED IUPAC NAME (VERIFY WITH YOUR OWN TRAIL)

butylcyclopropane

04 / THE ALKANE CHECK

EXPECTED IUPAC NAME: butylcyclopropane

The Alkane result: NOT YET CHECKED.

Insert genuine cropped tool screenshot after rebuilding the structure.

✓ parent checked ✓ locants checked ✓ formula counted ✓ name punctuated ☐ website check pending

CARD B01 — NAME → STRUCTURE

FOUNDATION • Given name: 2-methylpropane

01 / NAME DECODED

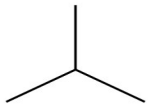
GIVEN NAME **2-methylpropane**

Parent	Propane (3 C)
Substituents	Methyl on C2
Expected formula	**C4H10**

03 / FORMULA CHECK + BUILD ORDER

Total carbons	3 + 1 = 4 C
Formula rule	Open chain: CnH2n+2 = C4H10
BUILD GUIDE	Draw a 3-carbon chain, then add one carbon branch to the middle carbon.

02 / REFERENCE STRUCTURE • THE ALKANE SCREENSHOT NEEDED



Study drawing only. Recreate it in The Alkane and add your real screenshot.

NOT a The Alkane capture | Original website check still pending

04 / BACK-CHECK

Expected: 2-methylpropane | Tool check: pending

✓ formula independently checked ☐ The Alkane screenshot ☐ tool name recorded

CARD B02 — NAME → STRUCTURE

FOUNDATION • Given name: 3-ethylpentane

01 / NAME DECODED

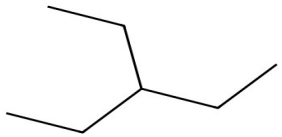
GIVEN NAME **3-ethylpentane**

Parent	Pentane (5 C)
Substituents	Ethyl on C3
Expected formula	**C7H16**

03 / FORMULA CHECK + BUILD ORDER

Total carbons	5 + 2 = 7 C
Formula rule	Open chain: CnH2n+2 = C7H16
BUILD GUIDE	Draw five parent carbons and add a two-carbon branch to carbon 3.

02 / REFERENCE STRUCTURE • THE ALKANE SCREENSHOT NEEDED



Study drawing only. Recreate it in The Alkane and add your real screenshot.

NOT a The Alkane capture | Original website check still pending

04 / BACK-CHECK

Expected: 3-ethylpentane | Tool check: pending

✓ formula independently checked ☐ The Alkane screenshot ☐ tool name recorded

CARD B03 — NAME → STRUCTURE

FOUNDATION • Given name: 2,2-dimethylbutane

01 / NAME DECODED

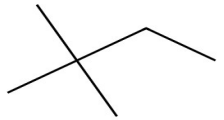
GIVEN NAME **2,2-dimethylbutane**

Parent	Butane (4 C)
Substituents	Two methyl groups on C2
Expected formula	**C6H14**

03 / FORMULA CHECK + BUILD ORDER

Total carbons	4 + 1 + 1 = 6 C
Formula rule	Open chain: CnH2n+2 = C6H14
BUILD GUIDE	Draw four parent carbons; connect two separate methyl branches to C2.

02 / REFERENCE STRUCTURE • THE ALKANE SCREENSHOT NEEDED



Study drawing only. Recreate it in The Alkane and add your real screenshot.

NOT a The Alkane capture | Original website check still pending

04 / BACK-CHECK

Expected: 2,2-dimethylbutane | Tool check: pending

✓ formula independently checked ☐ The Alkane screenshot ☐ tool name recorded

CARD B04 — NAME → STRUCTURE

DEVELOPING • Given name: 3-ethyl-2-methylpentane

01 / NAME DECODED

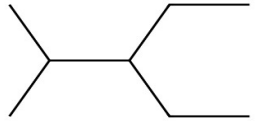
GIVEN NAME **3-ethyl-2-methylpentane**

Parent	Pentane (5 C)
Substituents	Ethyl on C3; methyl on C2
Expected formula	**C8H18**

03 / FORMULA CHECK + BUILD ORDER

Total carbons	5 + 2 + 1 = 8 C
Formula rule	Open chain: CnH2n+2 = C8H18
BUILD GUIDE	Draw pentane; put methyl at C2 and the two-carbon ethyl at C3.

02 / REFERENCE STRUCTURE • THE ALKANE SCREENSHOT NEEDED



Study drawing only. Recreate it in The Alkane and add your real screenshot.

NOT a The Alkane capture | Original website check still pending

04 / BACK-CHECK

Expected: 3-ethyl-2-methylpentane | Tool check: pending

✓ formula independently checked ☐ The Alkane screenshot ☐ tool name recorded

CARD B05 — NAME → STRUCTURE

DEVELOPING • Given name: 1-chloro-2-methylpropane

01 / NAME DECODED

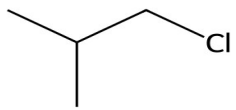
GIVEN NAME **1-chloro-2-methylpropane**

Parent	Propane (3 C)
Substituents	Chloro on C1; methyl on C2
Expected formula	**C4H9Cl**

03 / FORMULA CHECK + BUILD ORDER

Total carbons	3 + 1 = 4 C; 1 Cl
Formula rule	C4H10 − H + Cl = C4H9Cl
BUILD GUIDE	Draw propane; add methyl to carbon 2 and Cl to terminal carbon 1.

02 / REFERENCE STRUCTURE • THE ALKANE SCREENSHOT NEEDED



Study drawing only. Recreate it in The Alkane and add your real screenshot.

NOT a The Alkane capture | Original website check still pending

04 / BACK-CHECK

Expected: 1-chloro-2-methylpropane | Tool check: pending

✓ formula independently checked ☐ The Alkane screenshot ☐ tool name recorded

CARD B06 — NAME → STRUCTURE

DEVELOPING • Given name: 1,2-dimethylcyclopentane

01 / NAME DECODED

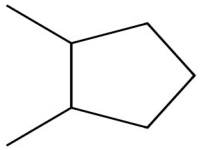
GIVEN NAME **1,2-dimethylcyclopentane**

Parent	Cyclopentane (5-C ring)
Substituents	Methyl groups at adjacent C1 and C2
Expected formula	**C7H14**

03 / FORMULA CHECK + BUILD ORDER

Total carbons	5 + 1 + 1 = 7 C
Formula rule	One ring: CnH2n = C7H14
BUILD GUIDE	Start with a 5-membered single-bond ring; attach methyl to adjacent ring vertices.

02 / REFERENCE STRUCTURE • THE ALKANE SCREENSHOT NEEDED



Study drawing only. Recreate it in The Alkane and add your real screenshot.

NOT a The Alkane capture | Original website check still pending

04 / BACK-CHECK

Expected: 1,2-dimethylcyclopentane | Tool check: pending

✓ formula independently checked ☐ The Alkane screenshot ☐ tool name recorded

CARD B07 — NAME → STRUCTURE

DEVELOPING • Given name: 4-ethyl-3,3-dimethylheptane

01 / NAME DECODED

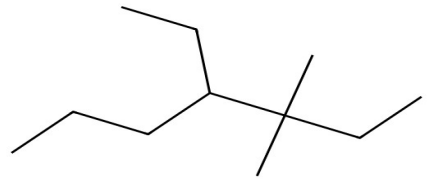
GIVEN NAME **4-ethyl-3,3-dimethylheptane**

Parent	Heptane (7 C)
Substituents	Ethyl on C4; two methyl groups on C3
Expected formula	**C11H24**

03 / FORMULA CHECK + BUILD ORDER

Total carbons	7 + 2 + 1 + 1 = 11 C
Formula rule	Open chain: CnH2n+2 = C11H24
BUILD GUIDE	Draw seven parent carbons; add two methyl at C3 and an ethyl at C4.

02 / REFERENCE STRUCTURE • THE ALKANE SCREENSHOT NEEDED



Study drawing only. Recreate it in The Alkane and add your real screenshot.

NOT a The Alkane capture | Original website check still pending

04 / BACK-CHECK

Expected: 4-ethyl-3,3-dimethylheptane | Tool check: pending

✓ formula independently checked ☐ The Alkane screenshot ☐ tool name recorded

CARD B08 — NAME → STRUCTURE

CHALLENGE • Given name: tert-butylcyclohexane

01 / NAME DECODED

GIVEN NAME **tert-butylcyclohexane**

Parent Cyclohexane (6-C ring)

Substituents tert-Butyl on C1 (no locant needed)

Expected formula **C10H20**

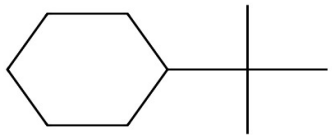
03 / FORMULA CHECK + BUILD ORDER

Total carbons 6 + 4 = 10 C

Formula rule One ring: CnH2n = C10H20

BUILD GUIDE Draw the 6-ring; attach a carbon directly to the ring, then three methyl groups to that carbon.

02 / REFERENCE STRUCTURE • THE ALKANE SCREENSHOT NEEDED



Study drawing only. Recreate it in The Alkane and add your real screenshot.

NOT a The Alkane capture | Original website check still pending

04 / BACK-CHECK

Expected: tert-butylcyclohexane | Tool check: pending

✓ formula independently checked ☐ The Alkane screenshot ☐ tool name recorded

CARD B09 — NAME → STRUCTURE

CHALLENGE • Given name: 3-ethyl-1,1-dimethylcyclohexane

01 / NAME DECODED

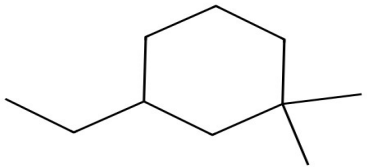
GIVEN NAME **3-ethyl-1,1-dimethylcyclohexane**

Parent	Cyclohexane (6-C ring)
Substituents	Two methyl groups on C1; ethyl on C3
Expected formula	**C10H20**

03 / FORMULA CHECK + BUILD ORDER

Total carbons	6 + 2 + 1 + 1 = 10 C
Formula rule	One ring: CnH2n = C10H20
BUILD GUIDE	Draw a 6-ring; attach two methyls at the same ring carbon and ethyl two positions away.

02 / REFERENCE STRUCTURE • THE ALKANE SCREENSHOT NEEDED



Study drawing only. Recreate it in The Alkane and add your real screenshot.

NOT a The Alkane capture | Original website check still pending

04 / BACK-CHECK

Expected: 3-ethyl-1,1-dimethylcyclohexane | Tool check: pending

✓ formula independently checked ☐ The Alkane screenshot ☐ tool name recorded

CARD B10 — NAME → STRUCTURE

CHALLENGE • Given name: 2-ethylbutane

01 / NAME DECODED

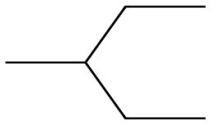
GIVEN NAME **2-ethylbutane**

Parent	Butane as WRITTEN (4 C)
Substituents	Ethyl on C2 as WRITTEN
Expected formula	**C6H14**

03 / FORMULA CHECK + BUILD ORDER

Total carbons	4 + 2 = 6 C
Formula rule	Open chain: CnH2n+2 = C6H14
BUILD GUIDE	Draw four carbons with a two-carbon ethyl on C2. Then trace the longest path.

02 / REFERENCE STRUCTURE • THE ALKANE SCREENSHOT NEEDED



Study drawing only. Recreate it in The Alkane and add your real screenshot.

NOT a The Alkane capture | Original website check still pending

04 / BACK-CHECK

Expected: 3-methylpentane (B10 name invalid) | Tool check: pending

RULE: 2-ethylbutane violates longest-chain rule. Trace a 5-C chain → 3-methylpentane.

✓ formula independently checked ☐ The Alkane screenshot ☐ tool name recorded